

State of Developer Ecosystem 2024: Language Trends, Satisfaction, and Strategic Shifts

JetBrains' annual survey of 23,262 developers
worldwide.

A comprehensive snapshot of
languages, tools, AI attitudes, and
career trends.

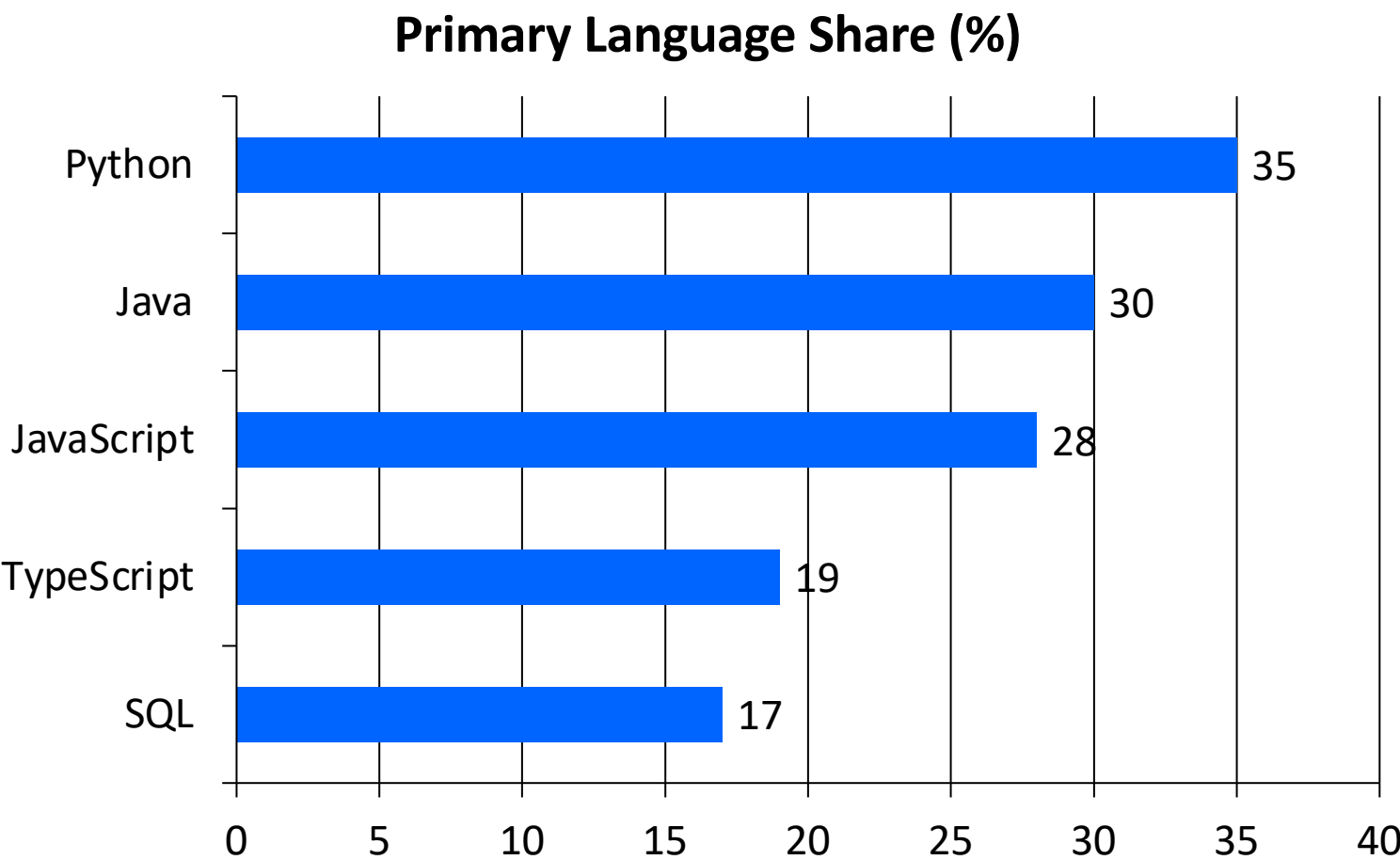
Python Closes the Gap: 8 Years of Language Usage Shifts

Language	2017	2020	2022	2024	Trend
JavaScript	65%	70%	65%	61%	Declining from peak
Python	32%	55%	53%	57%	Steady growth
HTML/CSS	60%	61%	54%	51%	Gradual decline
SQL	42%	56%	49%	48%	Plateaued
Java	47%	54%	48%	46%	Slow decline
TypeScript	12%	28%	34%	37%	Strong growth
Go	8%	19%	19%	18%	Plateaued after growth
Rust	--	7%	9%	11%	Steady growth
PHP	30%	27%	20%	17%	Significant decline
Kotlin	2%	17%	16%	14%	Plateaued

Key Takeaways:

JavaScript peaked at 70% in 2020 before declining to 61%.
Python rose from 32% to 57%, closing the gap.
TypeScript tripled from 12% to 37%.
PHP dropped from 30% to 17%.

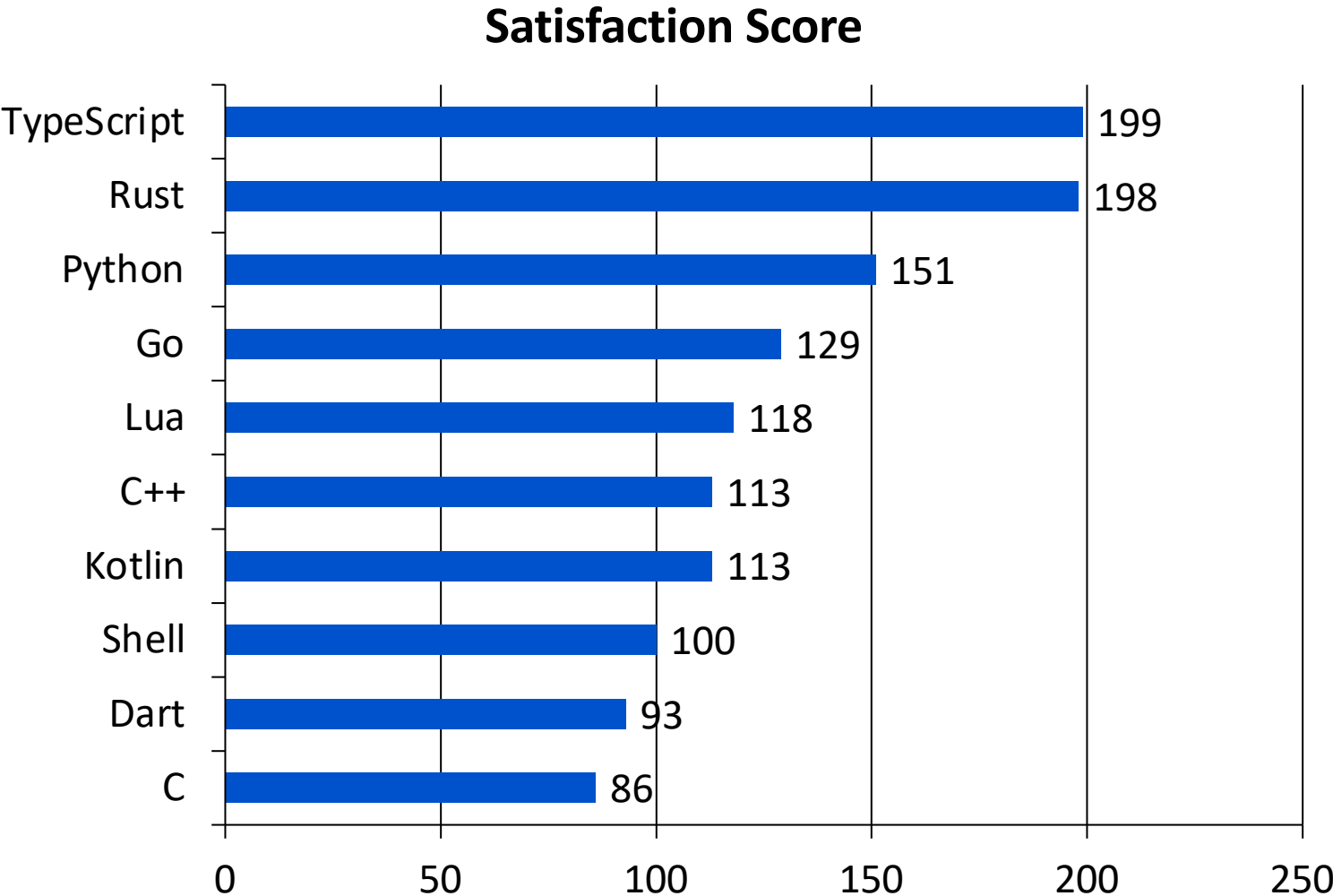
Python Is Now the #1 Primary Language at 35%



Ecosystem Lock-in:

Despite high primary usage rates, adoption intent for new primary languages remains below 6% across the board. This illustrates a strong 'comfort zone' effect where developers rarely plan to switch their primary stack.

TypeScript and Rust Top Developer Satisfaction, Nearly Tied at 199 vs 198



Key Insight:
Type-safe, modern-tooling languages dominate the top spots. The strong showing by TypeScript and Rust reflects developer enthusiasm for languages that combine safety and expressive type systems.

Each Language Owns a Domain: AI Belongs to Python, UI to TypeScript

Domain	Top Language	Share
AI / Machine Learning	Python	31%
Data Processing & Analytics	Python	17%
User Interface	TypeScript	62%
APIs & Services	Java	49%
Application Logic & Workflows	Java	55%
System Tools & Components	C++	27%
Integrating with APIs & Services	Python	38%

Language choice is heavily driven by domain requirements and ecosystem dominance, not general preference.

What This Means for Your Technology Strategy in 2025

- Invest in Python for AI/ML teams — it is the undisputed leader.
- TypeScript is the future of frontend and full-stack.
- Rust and Go are the growth plays for infrastructure and systems.
- PHP and Ruby hiring pools are shrinking.
- Developer satisfaction correlates with type safety and modern tooling — factor this into retention strategy.